This Presentation is Classified

UNCLASSIFIED//REL TO USA, FVEY

Disclosure Agreement

This brief has been approved for disclosure at the UNCLASSIFIED level for the countries of Australia, Canada, New Zealand, United Kingdom, United States. It is not the intention of the U.S. Marine Corps or the U.S. Government to provide training, skills, or instruction without the proper coordination per the governing U.S. Laws and Department of Defense policy and directives.

This information is furnished upon the condition that it or the knowledge of its possession, will not be released to another nation without the specific authority from the U.S. Marine Corps; that it will not be used for other than military purposes; that individual or corporate rights originating in the information, whether patented or not, will be respected; and that the information will be provided the same degree of security afforded it by the Department of Defense of the U.S. Regardless of any other markings on this document, it may not be declassified or downgraded without the written approval of the originating U.S. Agency.

This Presentation is Classified

UNCLASSIFIED

Fixed-wing Landing Zone: Criteria

GySgt J. W. “Chad” Kamphuis

Purpose

- The purpose of this course is to instruct prospective graduates on required criteria for a Landing Zone to be considered usable during Survey and Assault Zone Assessment (SAAZA).

Learning Objectives

- Identify the dimensions and gradients of the actual surfaces to include runways, overruns, and Zone A & B
- Identify the dimensions of the imaginary surfaces, to include slope ratio
- Define Reportable Region
- Define Controlling Region

References

- MMT TACSOP
- UFC 3-260-1 Airfield and Heliport Planning and Design
- TM 3-34.48-2 Theater of Operations: Roads, Airfields, and Heliports- Airfield and Heliport Design
- DAFMAN 13-217 Nuclear, Space, Missile, Command and Control; Drop Zone Landing Zone and Helicopter Landing Zone Procedures

References

- Engineering Technical Letter (ETL) 09-6 (17 Aug 09)
- TSPWG 3-260-03.02-19 Tri-Service Pavements Working Group (TSPWG) Manual. Airfield Pavement Evaluation Standards and Procedures
- Individual T/M/S NATOPS

Outline

- Introduction / Overview
- Unit One: Mathematics and Measurement
- Unit Two: Criteria
- Unit Three: Soil Identification and Evaluation
- Unit Four: SAAZA Process
- Unit Five: SAAZA Submission

Outline

- Assault Landing Zone (ALZ)
- Review
- Summary

Outline

- Assault Landing Zone (ALZ)
- Review
- Summary

Outline

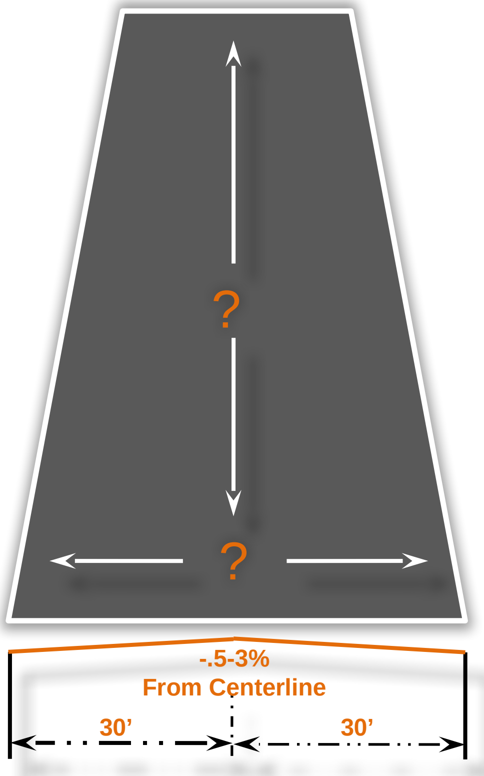
- Assault Landing Zone (ALZ)
  - Actual surfaces
  - Imaginary surfaces
- Review
- Summary

Outline

- Assault Landing Zone (ALZ)
  - Actual surfaces
  - Imaginary surfaces
- Review
- Summary

Actual Surfaces

- Runway
  - Rectangular area of an airfield prepared for the landing and takeoff of aircraft

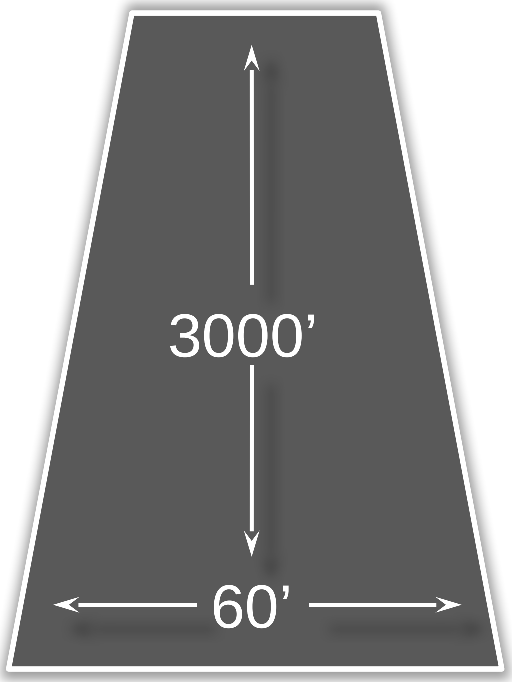


Actual Surfaces

- Actual Surfaces
  - Ground surfaces that an aircraft will operate on or near that will affect the aircraft's safety
    - Runway
    - Overrun
    - Zone A Vertical Obstruction Clearance
    - Zone B Vertical Obstruction Clearance

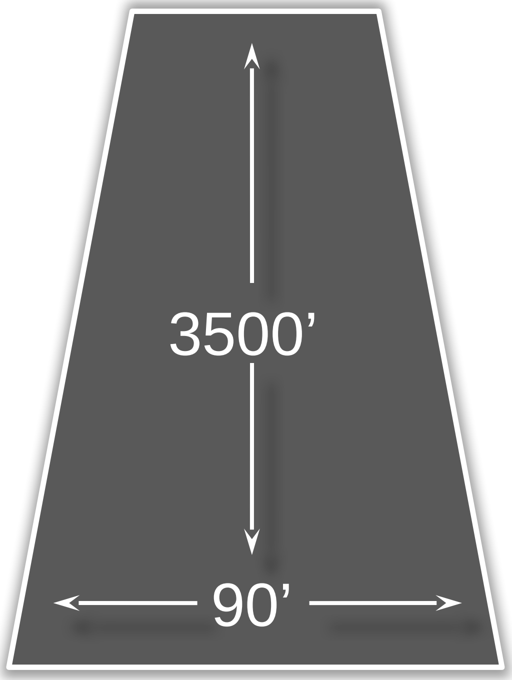
Actual Surfaces

- C-130 Semi-prepared Runway
  - Length = 3,500 ft
  - Width = 60 ft



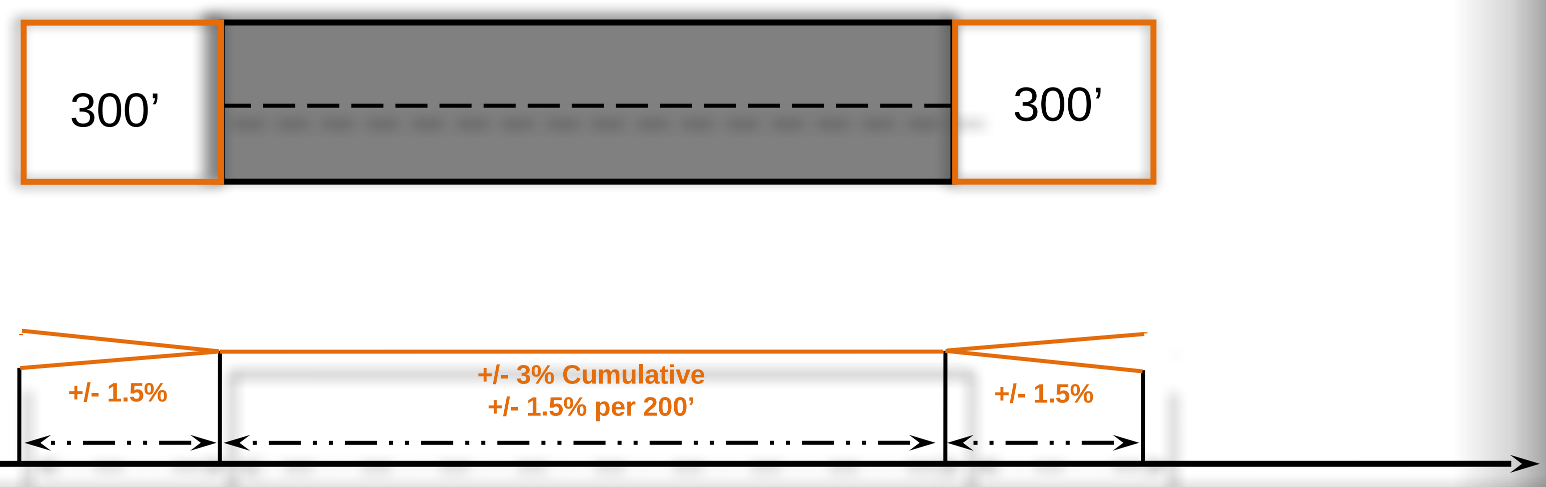
Actual Surfaces

- C-17 Semi-prepared Runway
  - Length = 3,500 ft
  - Width = 90 ft



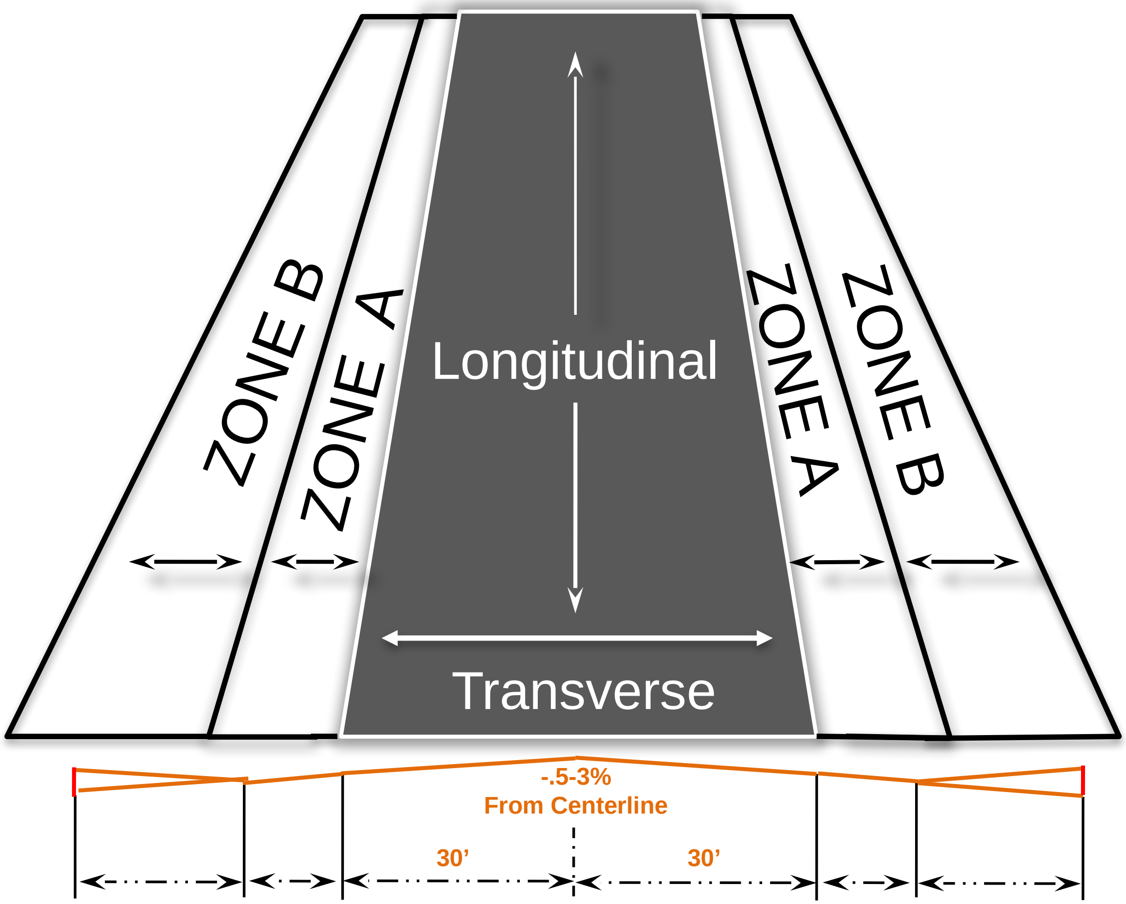
Actual Surfaces

- Overrun
  - An area the width of the runway, plus prepared shoulders, extending (300 ft) from the end of the runway into the clear zone



Actual Surfaces

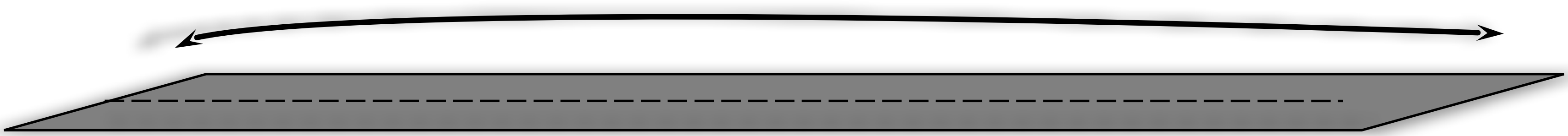
- Gradients
  - Longitudinal
  - Transverse



Actual Surfaces

Distance	Rise	Decimal Grade	Percent Grade
1-20	-4.5”	-.01875	-1.875%
21-40	-2”	-.00833	-.833%
41-60	-1”	-.0041	-.41%
61-80	+.5”	+.002	+.2%
81-100	+1.25”	+.0052	+.52%
101-120	-1.75”	-.0072	-.72%
121-140	-.5”	-.002	-.2%
141-160	-2.25”	-.0093	-.93%
161-180	-3.5”	-.0145	-1.45%
181-200	+.25”	-.001	-.1%
Cumulative	-13.5”	-.0056	-.56%

Longitudinal +/- 3%

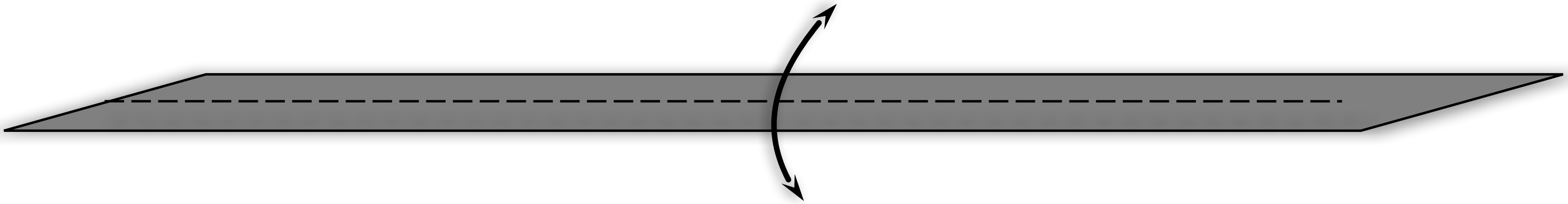


Maximum Longitudinal Change per 200 feet is +/- 1.5%

Actual Surfaces

Distance	Rise	Decimal Grade	Percent Grade
1-20	-4.5”	-.01875	-1.875%
21-40	-2”	-.00833	-.833%
41-60	-1”	-.0041	-.41%
61-80	+.5”	+.002	+.2%
81-100	+1.25”	+.0052	+.52%
101-120	-1.75”	-.0072	-.72%
121-140	-.5”	-.002	-.2%
141-160	-2.25”	-.0093	-.93%
161-180	-3.5”	-.0145	-1.45%
181-200	+.25”	-.001	-.1%
Cumulative	-13.5”	-.0056	-.56%

Transverse +/- .5% - 3%



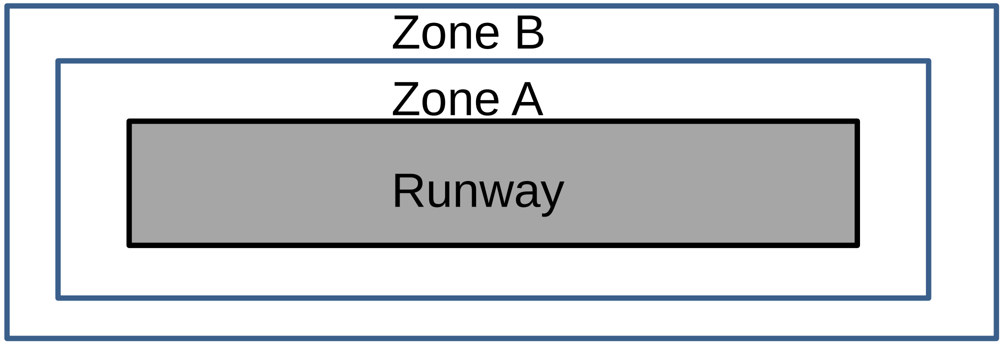
Actual Surfaces

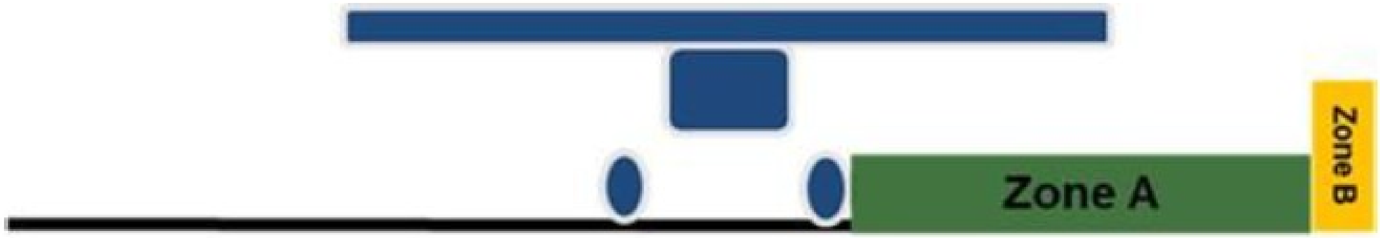
- Minimum Runway Dimensions
  – DAFMAN 13-217 Table 4.1

Type Aircraft	Length (Ft) (Note 1)	No Turn Required	180 Degree Turn (Normal)	180 Degree Turn (3 Point)	Taxiway
U-28/PC-12	2,000	30	30	30	23
MC-12	2,000	30	30	30	30
C-145	1,500	30/23	23	23	18
C-146	2,000	35/22	47.5	24.3	22
M/H/C-130H/J (**Note 2&5**)	3,000	60	60	50	30
C-130J-30 (**Note 2&5**)	3,000	60	85	75	30
C-17 (**Note 5**)	3,500	90	143	80	50

Actual Surfaces

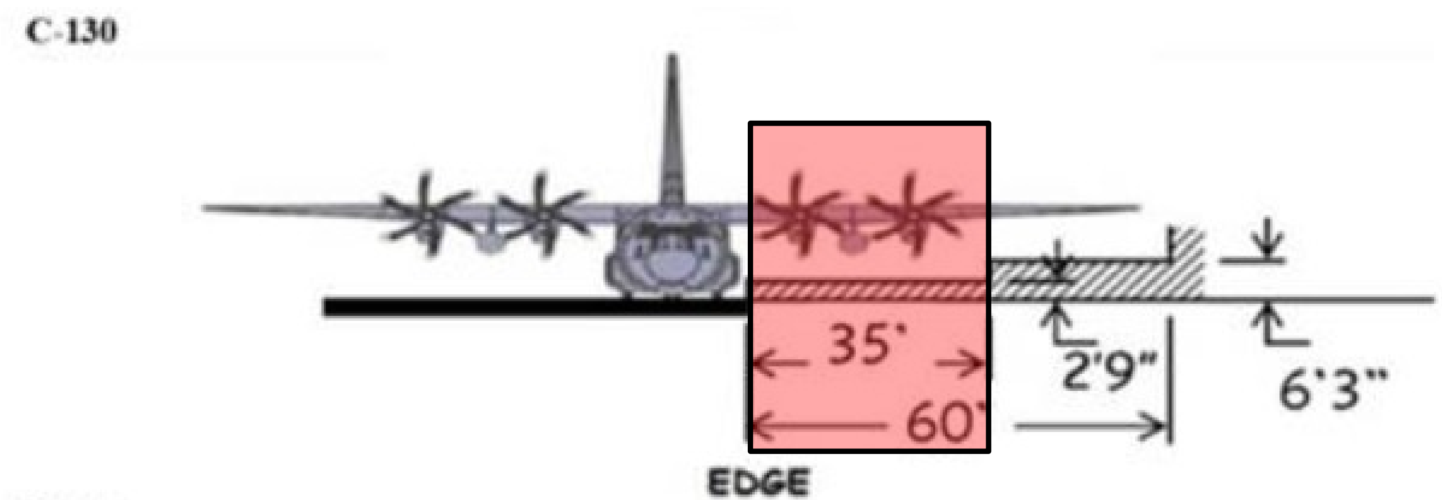
- Zones wrap around the entirety of the runway





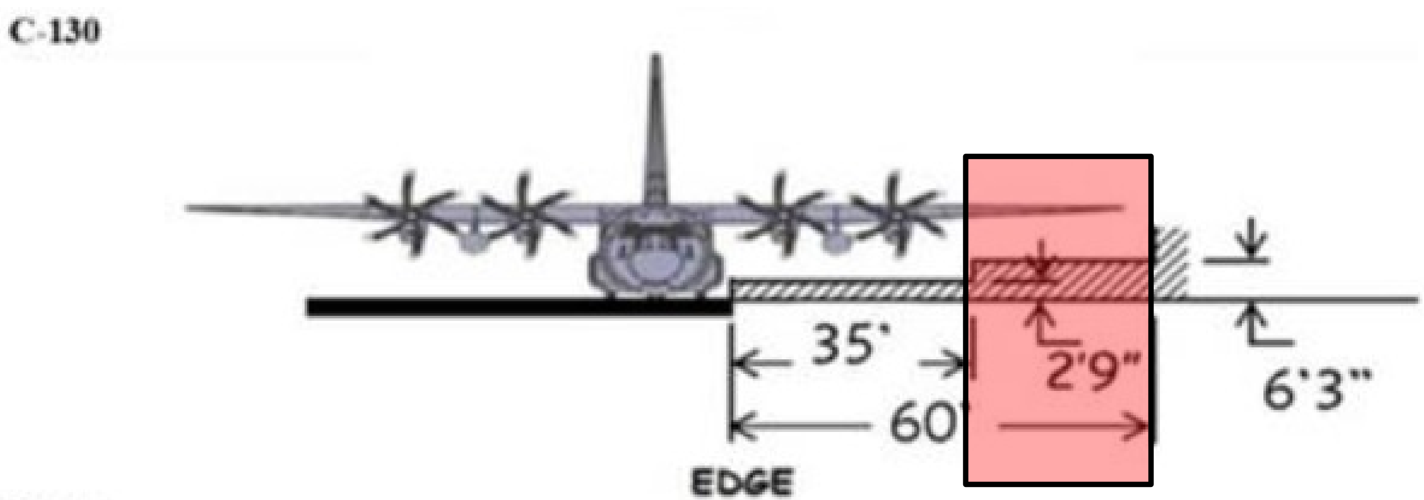
Actual Surfaces

- Zone A
  - Measure Zone A width from edge of useable runway
  - Zone A values account for potentially hazardous obstacles to the outboard nacelle



Actual Surfaces

- Zone B
  - Zone B extends outward from the outside edge of zone A.
  - Zone B values account for area from Zone A to the wing tip



Actual Surfaces

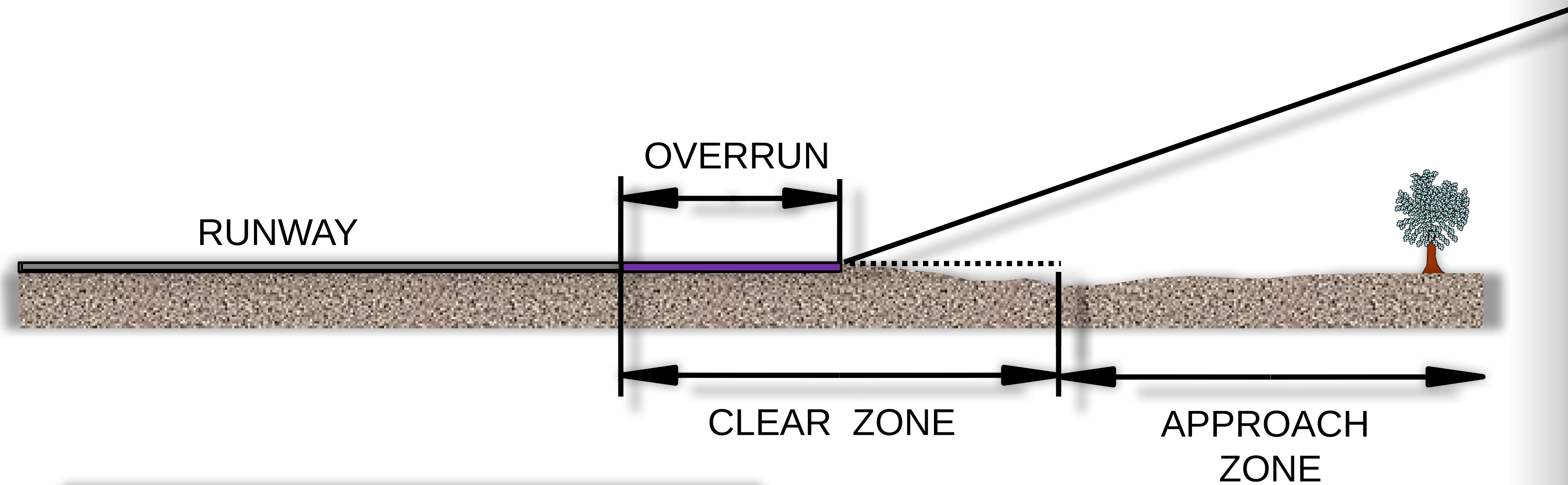
- AF Form 3822 may be utilized for evaluations of new LZ construction; when utilized, enter data for graded area and shoulders in blocks “B-Zone” and “A-Zone” respectively.

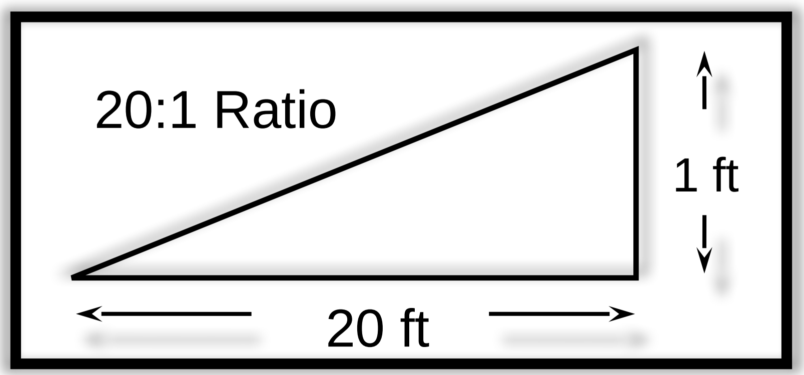
Imaginary Surfaces

- Imaginary Surfaces
  - Surfaces in space established around an LZ in relation to runways / helipads
  - Designed to protect airspace around the airfield
    - LZ Approach/Departure Vertical Obstruction Clearance
      - Reportable Region
      - Controlling Region

Imaginary Surfaces

- LZ Approach/Departure Vertical Obstruction Clearances

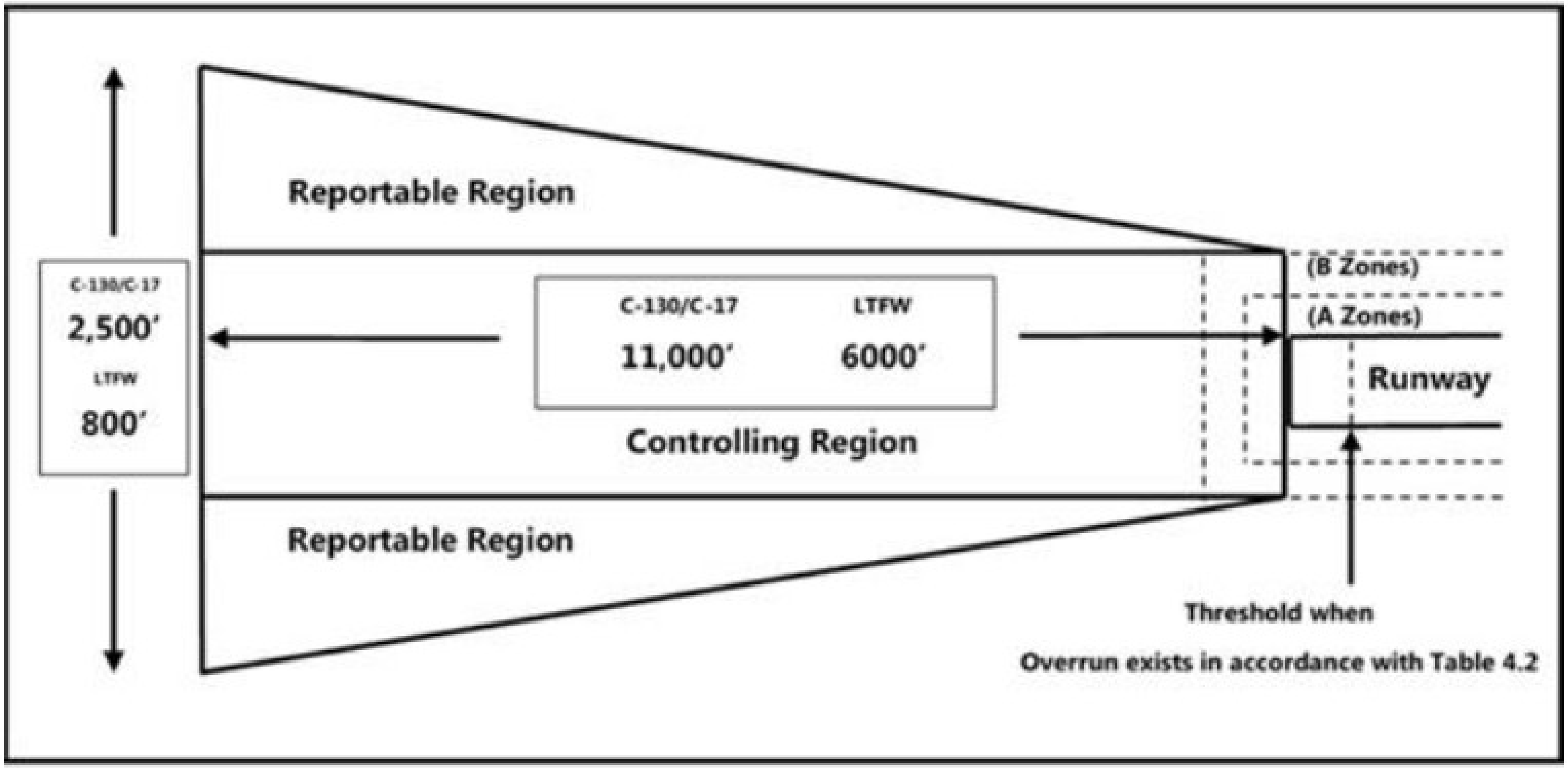




20:1 Slope for C-17

35:1 Slope for C-130

Imaginary Surfaces

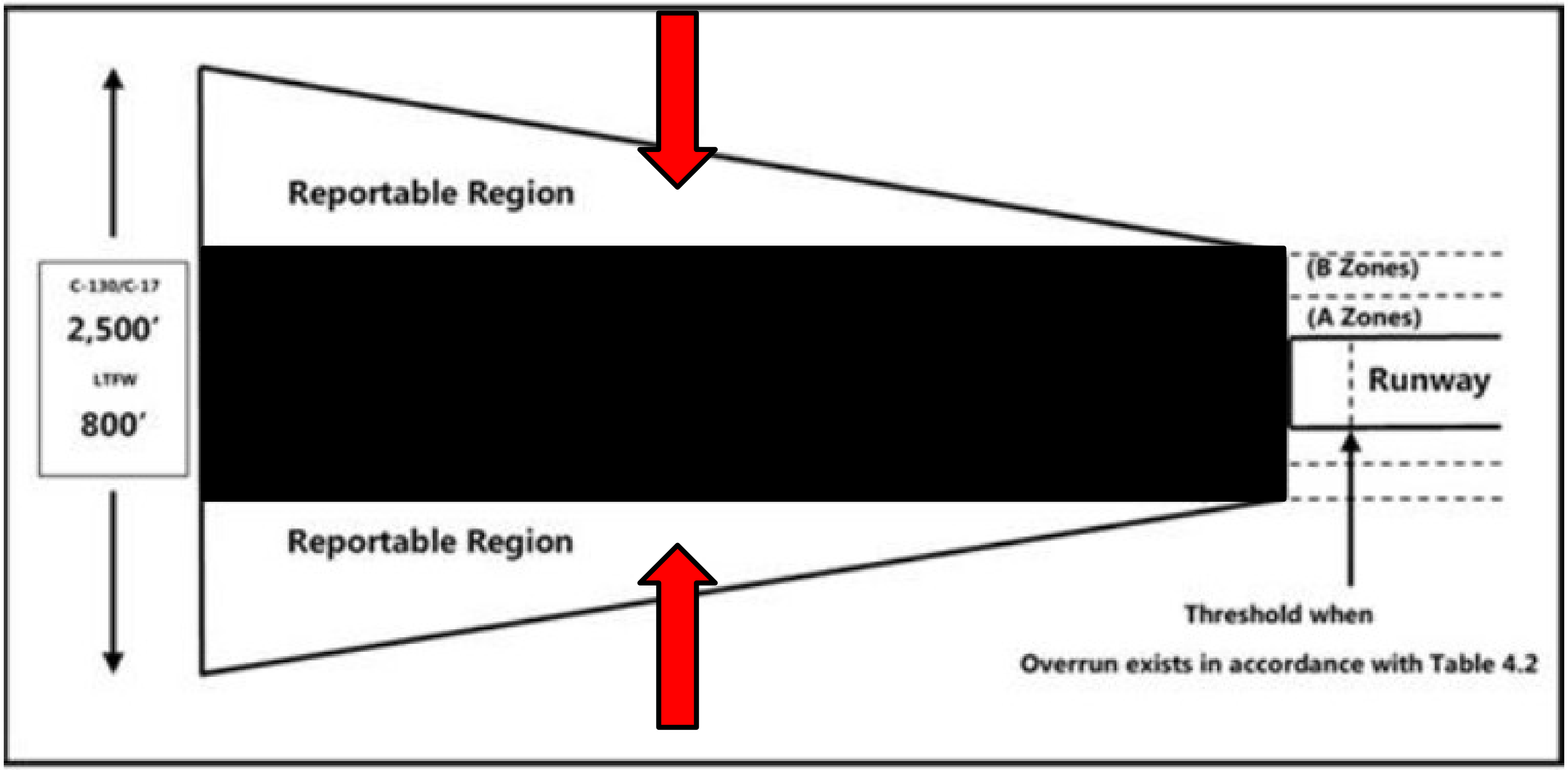


Imaginary Surfaces

- Reportable Region (RR)
  - Any penetration of Glide Slope Ratios (GSR) within the Approach-Departure Clearance Surface
  - The inner edge originates at the end of overrun or end of usable along the entirety of the width to the outside corners of B Zones
  - Obstacles within the RR do not drive the controlling obstacle or GSR to the runway

Imaginary Surfaces

- Reportable Region

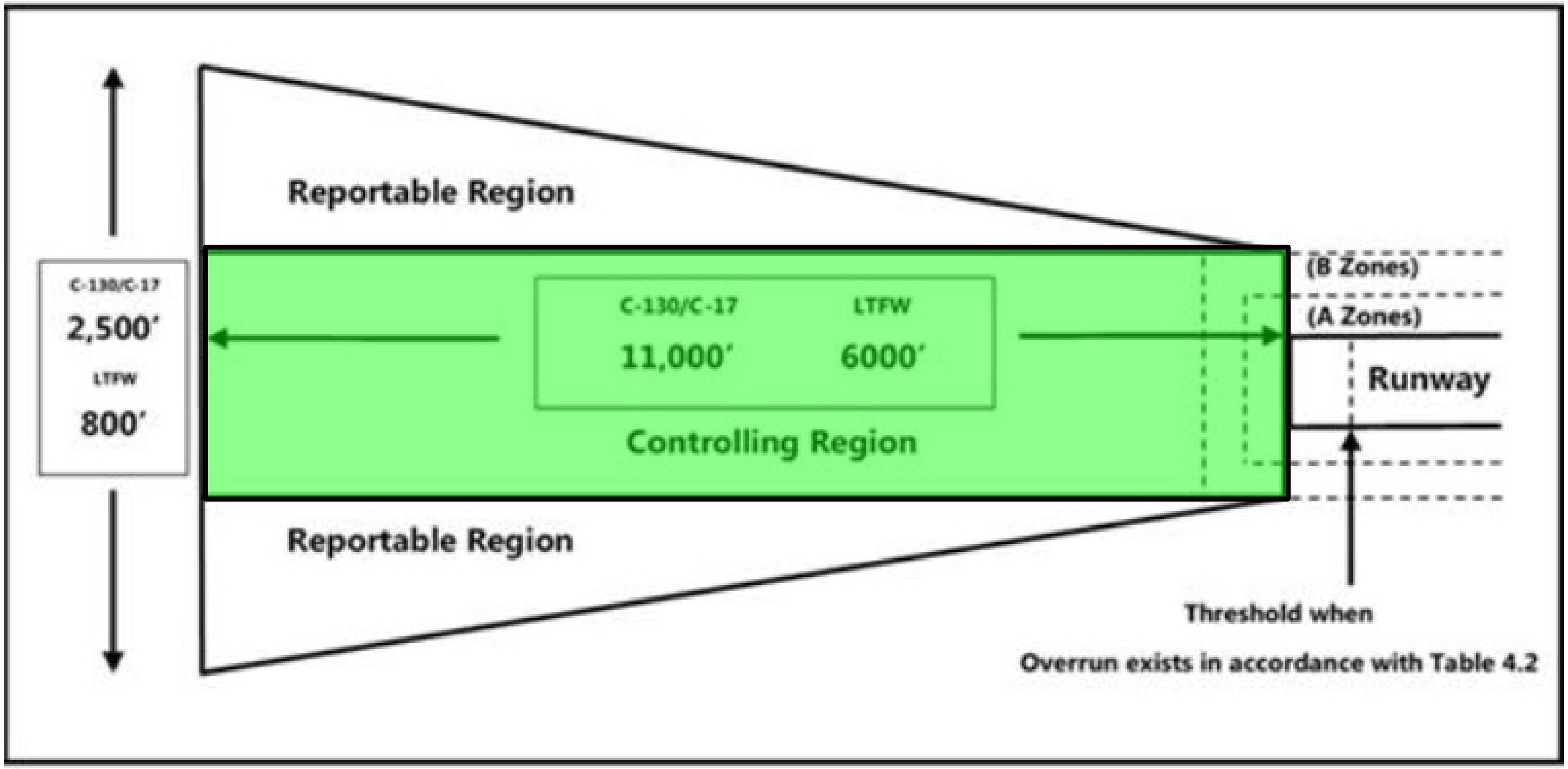


Imaginary Surfaces

- Controlling Region (CR)
  - Extends from the runway threshold to the end of the approach departure clearance surface (T/M/S specific)
  - Controlling GSR is derived from the most restrictive obstacle (controlling obstacle) within the CR
  - Controlling GSR for at least one runway must meet the minimum departure obstacle clearance requirement

Imaginary Surfaces

- Controlling Region



Actual/Imaginary Surfaces

- LZ Runway Slopes, Overruns, and Vertical Obstruction Clearances
  - DAFMAN 13-217 Table 4.2

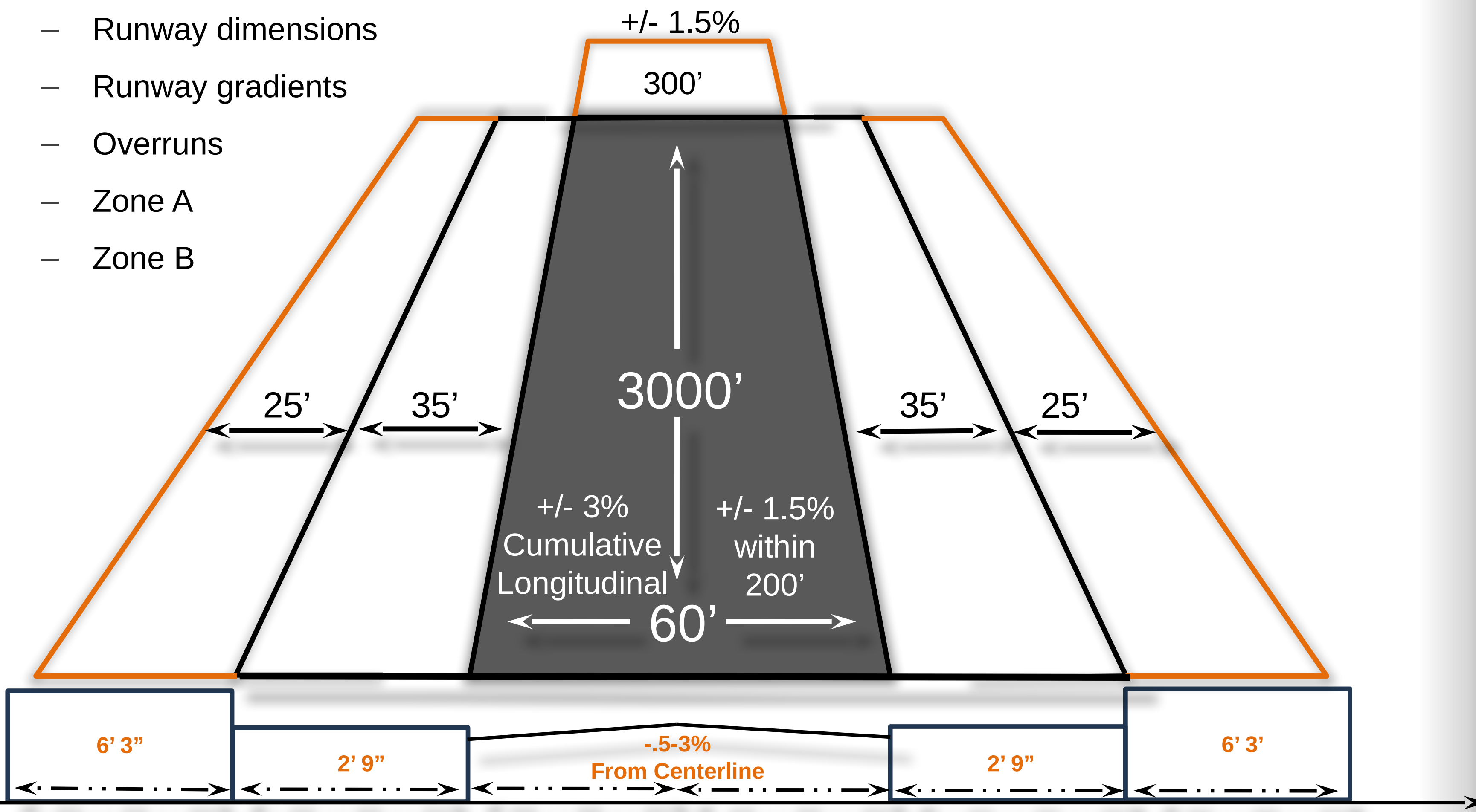
Type AC	Rwy length slope (Note 1) Rwy slope change (Note 1) Rwy width slope (Note 1)	Rwy Overrun Length (Note 2) Zone A Width/Height (Note 3) Zone B Width/Height (Note 3)	Glide Slope Ratio (GSR) (Note 4)
U-28/PC-12/ MC-12	± 4% longitudinal Refer to operation manuals ± 3% transverse	0’ minimum, 100’ when available 19’ 3”, no obstacle higher than 4” 5’ 4”, no obstacle higher than 5’	12:1 (4.76°)
C-130	± 3% longitudinal Refer to operation manual ± 0.5-3% transverse	300’ minimum 35’, no obstacle higher than 2’ 9” 25’, no obstacle higher than 6’ 3”	35:1 (1.64°)
C-17	± 2% longitudinal Refer to operation manual ± 0.5-3% transverse	300’ minimum 35’, no obstacle higher than 3’ 33’, no obstacle higher than 5’	20:1 (2.86°)
C-145A	± 6% longitudinal Refer to operation manual ± 4% transverse	0’ minimum, 100’ when available 30’ 7”, no obstacle higher than 3’ 7’ 3”, no obstacle higher than 5’	8:1 (7.13°)
C-146A	± 2% longitudinal Refer to operation manual ± 3% transverse	0’ minimum, 100’ when available 28’ 11”, no obstacle higher than 3’ 6’ 11”, no obstacle higher than 5’	20:1 (2.86°)

Outline

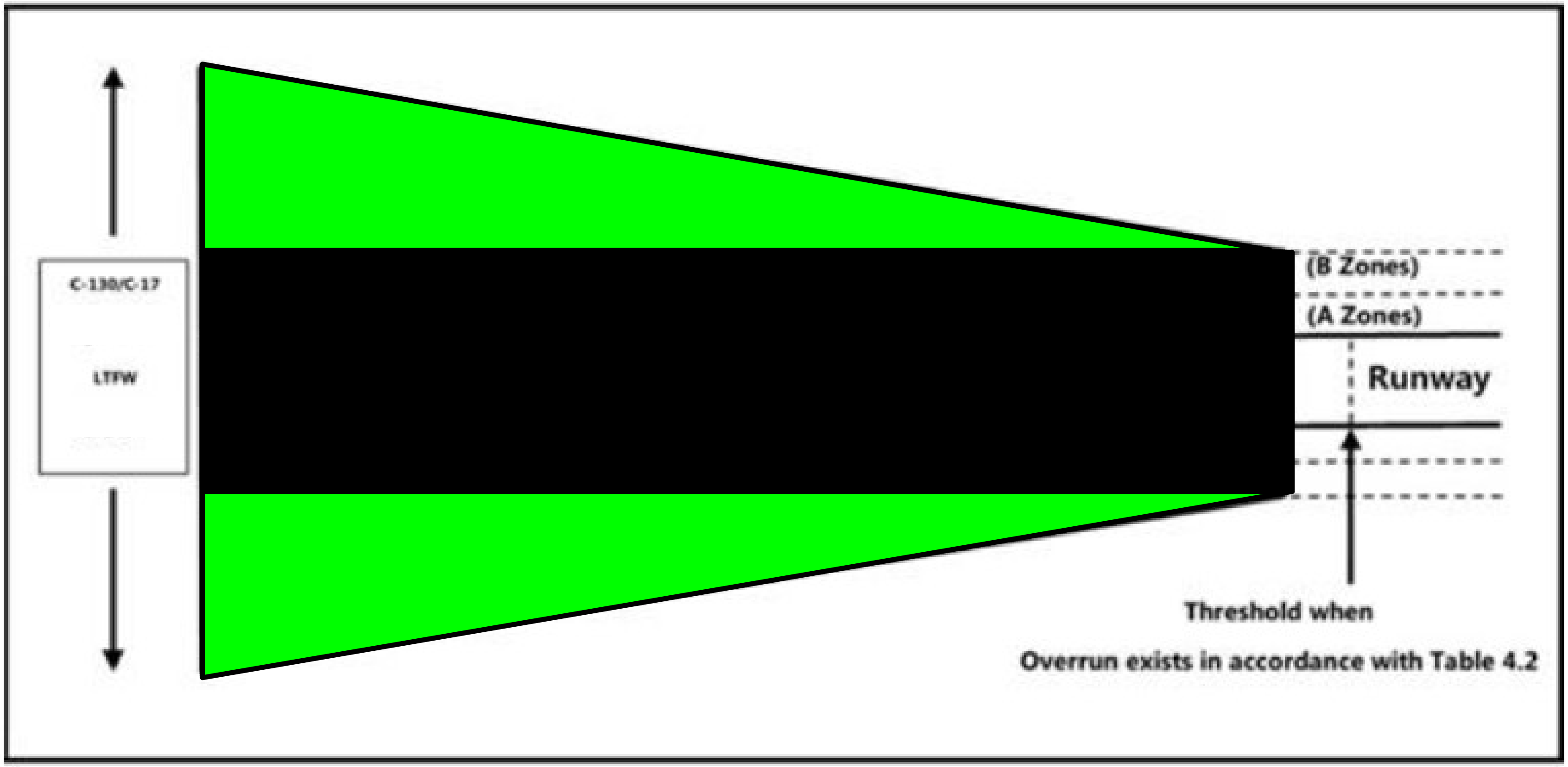
- Assault Landing Zone (ALZ)
- Review
- Summary

Review

- Identify the dimensions and gradients of the runway minimums for a C130, to include:
  - Runway dimensions
  - Runway gradients
  - Overruns
  - Zone A
  - Zone B



Review


C-130/C-17
LTFW
(B Zones)
(A Zones)
Runway
Threshold when
Overrun exists in accordance with Table 4.2

Outline

- Assault Landing Zone (ALZ)
- Review
- Summary

Questions / Critiques

This Presentation is Classified

UNCLASSIFIED